

A Coupled H^{-1} Gradient Flow

1 Definitions

Let

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}).$$

For real periodic functions u and v , define

$$\mu_u = -u_{xx} + u(u^2 + v^2) + (u - v), \quad \mu_v = -v_{xx} + v(u^2 + v^2) + (v - u). \quad (1)$$

Consider

$$u_t = \partial_x^2 \mu_u, \quad v_t = \partial_x^2 \mu_v, \quad (u, v)|_{t=0} = (u_0, v_0), \quad (2)$$

where

$$u_0, v_0 \in H^2(\mathbb{T}), \quad \int_{\mathbb{T}} u_0 dx = \int_{\mathbb{T}} v_0 dx = 0. \quad (3)$$

For every integrable periodic function $g \in L^1(\mathbb{T})$, the angular brackets denote its normalized spatial average over $\mathbb{T}$:

$$\langle g \rangle := \frac{1}{|\mathbb{T}|} \int_{\mathbb{T}} g(x) dx = \frac{1}{2\pi} \int_{\mathbb{T}} g(x) dx, \quad |\mathbb{T}| = 2\pi. \quad (4)$$

Using this notation, define

$$\mathcal{E}(u, v) = \left\langle \frac{u_x^2 + v_x^2}{2} + \frac{(u^2 + v^2)^2}{4} + \frac{(u - v)^2}{2} \right\rangle. \quad (5)$$

2 Problem Formalization

Prove all of the following assertions, or identify the first false assertion and give a counterexample.

1. There is a unique global solution satisfying

$$(u, v) \in C([0, \infty); H^2(\mathbb{T}))^2 \cap L_{\text{loc}}^2([0, \infty); H^4(\mathbb{T}))^2, \quad (6)$$

and

$$(u_t, v_t) \in L_{\text{loc}}^2([0, \infty); L^2(\mathbb{T}))^2. \quad (7)$$

2. On every bounded interval $[0, T]$, the solution map is locally Lipschitz from $H^2 \times H^2$ into

$$C([0, T]; H^2 \times H^2). \quad (8)$$

3. For every $t \geq 0$,

$$\mathcal{E}(t) + \int_0^t \langle |\partial_x \mu_u|^2 + |\partial_x \mu_v|^2 \rangle d\tau = \mathcal{E}(0). \quad (9)$$

4. The prefactor-one estimate

$$\mathcal{E}(t) \leq e^{-2t} \mathcal{E}(0) \quad (10)$$

holds, and the exponent 2 cannot be uniformly increased.

5. For $t \geq 1$,

$$\|u(t)\|_{H^2} + \|v(t)\|_{H^2} \leq C(u_0, v_0) e^{-t}. \quad (11)$$